

Year 8 Mini Test 4

Geometry - Congruent Triangles, transformations and angle properties.

Name : Solutions.

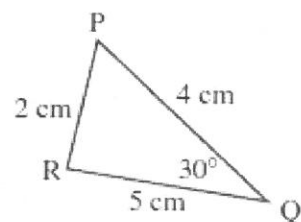
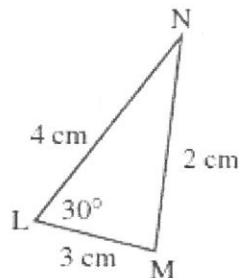
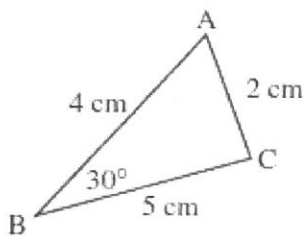
Score: 32.. / 35 Marks

Show working and answers on this sheet. Show ALL working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Time allocated **35 mins**

1. For each set of three triangles below, determine which of the three triangles are congruent, state which rule proves they are congruent and write a statement of congruence. (Marks = 3,3,3,3)

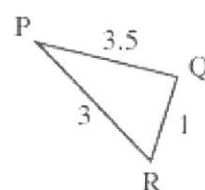
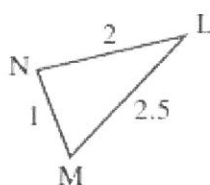
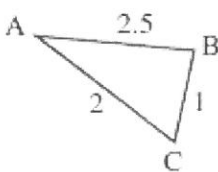
a



$\triangle ABC \equiv \triangle PQR$  (SAS) right order ✓

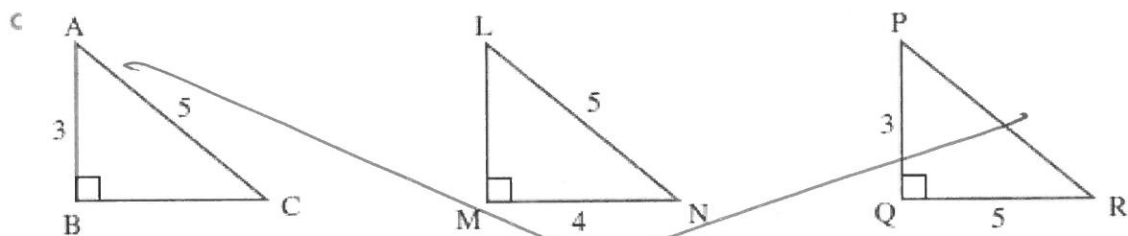
[3 Marks]

b

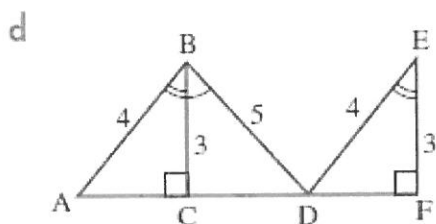


$\triangle ABC \equiv \triangle LNM$  (SSS) right order ✓

[3 Marks]



[3 Marks]



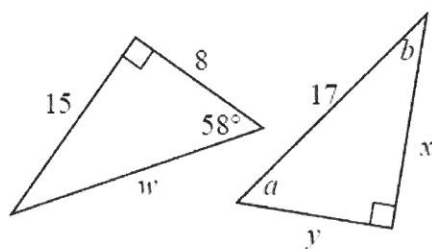
$$\triangle ABC \equiv \triangle DEF \text{ (RHS)}$$

right  
order

[3 Marks]

2. Given that each set of triangles below are congruent to each other, find the missing values.

A)

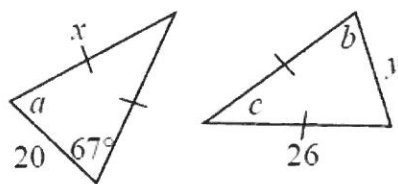


$$a = 58 \quad b = 32 \quad w = 17$$

$$x = 15 \quad y = 8$$

[5 Marks]

B)



$$a = 67 \quad b = 67 \quad c = 46$$

$$x = 26 \quad y = 20$$

[5 Marks]

$$67 + 67 = 134$$

3. Circle yes or no if these are a valid test for congruence between 2 triangles

1. All sides of are the same length on both triangles - Yes/No

2. All angles are the same size on both triangles - Yes/No

3. 2 sides and the angles between them are the same on both triangles - Yes/No

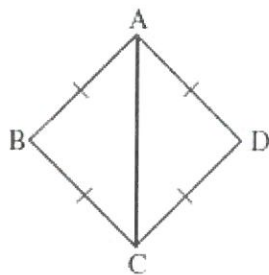
4. Both triangles have right angles, the hypotenuse is the same size and 1 other side is the same length. - Yes/No

5. 2 sides are the same length and 1 angle is the same size on both triangles - Yes/No

[5 marks]

4. Explain which rule of congruence proves that each of the following pictures is made from 2 congruent triangles.

a)



$\triangle ABC \cong \triangle ADC$

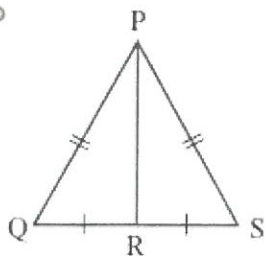
a)  $AB = AD$  (Given)

$BC = DC$  (Given)

$AC$  Common.  
(Side, side, side)

[2 marks]

b)



$\triangle PQR \cong \triangle PSR$

b)  $PQ = PS$

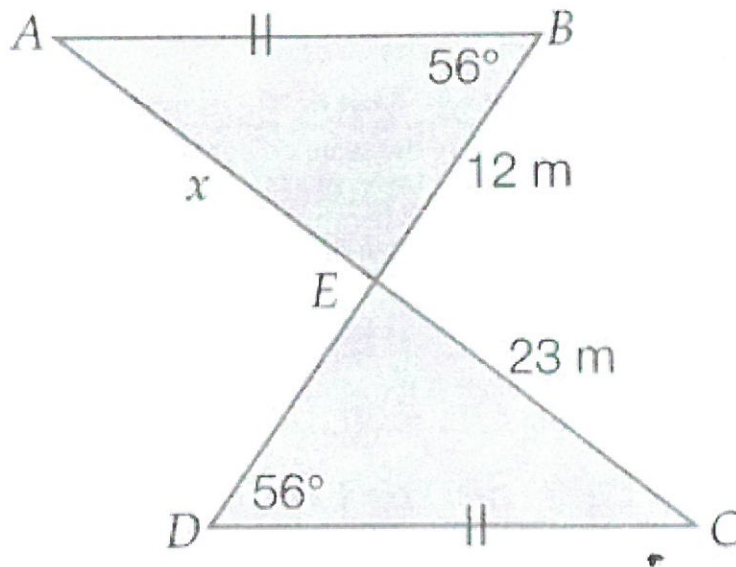
$QR = SR$

$PR$  (Common)

(SSS)

[2 marks]

5. Prove that the pair of triangles below is congruent and hence find the value of the value marked with an  $x$ .



~~$\triangle ABE$~~   $AB = DC$  (Given) ✓

$\angle AEB = \angle CED$  (vertically opp) ✓

$\angle ABE = \angle EDC$  (Given) ✓

$\triangle ABE \equiv \triangle CDE$  (AAS) ✓

$x = 23\text{ m}$  ✓

[4 Marks]

End of test